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DEPARTMENT OF MATHEMATICS AND ACTUARIAL SCIENCE

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UNIT NAME: Elements of Linear Algebra

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Course Description

Vector spaces over $\mathbb{R}$. Linear independence. Matrices: properties, operations, determinants, systems of linear equations. Eigenvalues and eigenvectors. Quadratic forms. Orthogonal matrices. Matrix differentiation and maximization problems

References

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Vector spaces over $\mathbb{R}$

Vectors and Scalars

Vectors are quantities characterized by both magnitude and direction, such as displacement, velocity, force and acceleration.

To describe such quantities, we introduce the concept of a vector as a directed line segment $\overrightarrow{pq}$ from one point called the initial point, p, to another point called the terminal point, q.

The magnitude or length of the vector $\vec{A}$ is denoted by $|\vec{A}|$.

Quantities that are characterized by magnitude only, such as mass, length and temperature are called scalars. A vector whose length or absolute value is one is called a unit vector. A vector is said to be zero if its absolute value is zero. Vectors of equal length which have the same direction are said to be equal.

Definition

By the product of a scalar λ and a vector $\vec{A}$ we mean the vector $\lambda\vec{A}$ whose length is equal to the product of $|\lambda|$ and the magnitude of $\vec{A}$ and whose direction is that of $\vec{A}$ if λ is positive and opposite to it if λ is negative.

Definition

A vector space over $\mathbb{R}$ is a nonempty set V of objects called vectors on which are defined two operations called addition and multiplication by scalars (real numbers) in accordance with the following axioms (rules):

- (i) If $\vec{v}_1, \vec{v}_2 \in V$, then $\vec{v}_1 + \vec{v}_2 \in V$
- (ii) If $\vec{v} \in V$ and $\lambda \in \mathbb{R}$, then $\lambda\vec{v} \in V$
- (iii) If $\vec{v}_1, \vec{v}_2, \vec{v}_3 \in V$, then $\vec{v}_1 + (\vec{v}_2 + \vec{v}_3) = (\vec{v}_1 + \vec{v}_2) + \vec{v}_3$
- (iv) If $\vec{v}_1, \vec{v}_2 \in V$, then $\vec{v}_1 + \vec{v}_2 = \vec{v}_2 + \vec{v}_1$
- (v) $\exists$ a vector in V denoted by $\vec{0}$ called the zero vector
- (vi) $\forall \vec{v} \in V \exists$ a unique vector $-\vec{v} \ni \vec{v} + (-\vec{v}) = \vec{0}$

- (vii) If $\lambda_1, \lambda_2 \in \mathbb{R}$ and $\vec{v} \in V$ then $\lambda_1(\lambda_2\vec{v}) = (\lambda_1\lambda_2)\vec{v} = \lambda_2(\lambda_1\vec{v})$
- (viii) If $\lambda_1, \lambda_2 \in \mathbb{R}$ and $\vec{v} \in V$ then $(\lambda_1 + \lambda_2)\vec{v} = \lambda_1\vec{v} + \lambda_2\vec{v}$
- (ix) If $\lambda \in \mathbb{R}$ and $\vec{v}_1, \vec{v}_2 \in V$, then $\lambda(\vec{v}_1 + \vec{v}_2) = \lambda\vec{v}_1 + \lambda\vec{v}_2$
- (x) If $\vec{v} \in V$ then $1 \cdot \vec{v} = \vec{v}$

Examples

1. $\mathbb{R}^n$ is a vector space with operations defined as follows:

(i) For $(a_1, a_2, \dots, a_n), (b_1, b_2, \dots, b_n) \in \mathbb{R}^n$ then

$$(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n) \in \mathbb{R}^n$$

(ii) For $\lambda \in \mathbb{R}$ and $(a_1, a_2, \dots, a_n) \in \mathbb{R}^n$ then

$$\lambda(a_1, a_2, \dots, a_n) = (\lambda a_1, \lambda a_2, \dots, \lambda a_n) \in \mathbb{R}^n$$

2. Let $V = \{\text{all } 2 \times 2 \text{ matrices with real entries}\}$. This is a vector space with operations defined as follows:

(i) For $\begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix}, \begin{pmatrix} b_1 & b_2 \\ b_3 & b_4 \end{pmatrix} \in V$ then $\begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} + \begin{pmatrix} b_1 & b_2 \\ b_3 & b_4 \end{pmatrix} = \begin{pmatrix} a_1 + b_1 & a_2 + b_2 \\ a_3 + b_3 & a_4 + b_4 \end{pmatrix} \in V$

(ii) For $\lambda \in \mathbb{R}$ and $\begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \in V$ then $\lambda \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} = \begin{pmatrix} \lambda a_1 & \lambda a_2 \\ \lambda a_3 & \lambda a_4 \end{pmatrix} \in V$

3. Let $V = \{\text{polynomials with real coefficients of degree } \leq 3\}$. This is a vector space. For if,

$$f(x) = 2 + x - 3x^2 + x^3 \in V$$

$$g(x) = 1 - 2x + 4x^2 + 3x^3 \in V$$

And $\lambda \in \mathbb{R}$, then

$$f(x) + g(x) = 3 - x + x^2 + 4x^3 \in V$$

$$\lambda f(x) = 2\lambda + \lambda x - 3\lambda x^2 + \lambda x^3 \in V$$

Vector subspaces

Definition

Let V be a vector space and $U \subseteq V$. Then U is a vector subspace of V if for

(a) $\vec{u}_1, \vec{u}_2 \in U$ then $\vec{u}_1 + \vec{u}_2 \in U$

(b) $\lambda \in \mathbb{R}$ and $\vec{u} \in U$ then $\lambda \vec{u} \in U$

Examples

1. Let $A = \{(x, y) : x = 0\} \subseteq \mathbb{R}^2$. This is the y-axis. To prove that A is a subspace of $\mathbb{R}^2$,

(i) Let

$$\begin{aligned}\vec{u} &= (x_1, y_1), \vec{v} = (x_2, y_2) \in A \\ &\Rightarrow x_1 = x_2 = 0\end{aligned}$$

$$\text{Therefore, } \vec{u} + \vec{v} = (0, y_1) + (0, y_2) = (0, y_1 + y_2) \in A$$

$$\text{So } \vec{u} + \vec{v} \in A$$

(ii) Let

$$\begin{aligned}\lambda \in \mathbb{R} \text{ and } \vec{u} &= (x, y) \in A \\ &\Rightarrow x = 0\end{aligned}$$

$$\text{Therefore, } \lambda \vec{u} = (\lambda x, \lambda y) = (0, \lambda y) \in A$$

$$\text{Thus, } \lambda \vec{u} \in A$$

Hence, A is a subspace of $\mathbb{R}^2$.

2. Let $B = \{(x, y) : x = 2y\} \subseteq \mathbb{R}^2$. This is a straight line passing through the origin of slope

$\frac{1}{2}$. To prove that B is a subspace of $\mathbb{R}^2$,

(i) Let

$$\begin{aligned}\vec{u} &= (x_1, y_1), \vec{v} = (x_2, y_2) \in B \\ &\Rightarrow x_1 = 2y_1 \text{ and } x_2 = 2y_2\end{aligned}$$

Therefore,

$$\begin{aligned}\vec{u} + \vec{v} &= (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \\ &\Rightarrow x_1 + x_2 = 2y_1 + 2y_2 = 2(y_1 + y_2) \\ &\Rightarrow \vec{u} + \vec{v} \in B\end{aligned}$$

(ii) Let

$$\begin{aligned}\lambda \in \mathbb{R} \text{ and } \vec{u} &= (x, y) \in B \\ &\Rightarrow x = 2y\end{aligned}$$

Therefore,

$$\begin{aligned}\lambda \vec{u} &= (\lambda x, \lambda y) \\ \Rightarrow \lambda x &= \lambda(2y) = 2\lambda y = 2(\lambda y)\end{aligned}$$

So, $\lambda \vec{u} \in B$

Hence, B is a vector subspace of $\mathbb{R}^2$.

3. Let $C = \{(x, y) : x^2 = y^2\} \subseteq \mathbb{R}^2$. This is a union of two straight lines passing through the origin. To check whether C is a vector subspace of $\mathbb{R}^2$,

Let

$$\vec{a} = (x, y), \vec{b} = (-x, y) \in C$$

Then,

$$\vec{a} + \vec{b} = (x, y) + (-x, y) = (0, 2y) \notin C$$

Thus, C is not a vector subspace of $\mathbb{R}^2$.

4. Let $\vec{a} \in \mathbb{R}^n$ and $U = \{\vec{u} \in \mathbb{R}^n : \vec{a} \cdot \vec{u} = 0\}$. Prove that U is a vector subspace of $\mathbb{R}^n$.

Proof

Take

$$\begin{aligned}\vec{u}_1, \vec{u}_2 &\in U \\ \Rightarrow \vec{a} \cdot \vec{u}_1 &= \vec{a} \cdot \vec{u}_2 = 0\end{aligned}$$

Hence,

$$\begin{aligned}\vec{a} \cdot (\vec{u}_1 + \vec{u}_2) &= \vec{a} \cdot \vec{u}_1 + \vec{a} \cdot \vec{u}_2 = 0 + 0 = 0 \\ \Rightarrow \vec{u}_1 + \vec{u}_2 &\in U\end{aligned}$$

Thus, the 1st condition is satisfied.

Let

$$\begin{aligned}\lambda \in \mathbb{R} \text{ and } \vec{u} &\in U \\ \Rightarrow \vec{a} \cdot \vec{u} &= 0\end{aligned}$$

We want to show that $\lambda \vec{u} \in U$.

Now,

$$\begin{aligned}\vec{a} \cdot \lambda \vec{u} &= \lambda (\vec{a} \cdot \vec{u}) = 0 \\ \Rightarrow \lambda \vec{u} &\in U\end{aligned}$$

Hence, the 2nd condition holds. Therefore, U is a vector subspace of $\mathbb{R}^n$.

Theorem

Let U_1, U_2 be subspaces of a vector space V . Then, if $U_1 \not\subseteq U_2$ and $U_2 \not\subseteq U_1$, then $U_1 \cup U_2$ is not a subspace of V .

Proof

Take $\vec{u}_1 \in U_1$, $\vec{u}_1 \notin U_2$ and $\vec{u}_2 \notin U_1$, $\vec{u}_2 \in U_2$.

Then, $\vec{u}_1, \vec{u}_2 \in U_1 \cup U_2$.

Assume

$$\begin{aligned}\vec{u}_1 + \vec{u}_2 &\in U_1 \\ \Rightarrow (\vec{u}_1 + \vec{u}_2) - \vec{u}_1 &\in U_1 \\ \Rightarrow \vec{u}_2 &\in U_1 \\ \Rightarrow a \text{ contradiction} \\ \Rightarrow \vec{u}_1 + \vec{u}_2 &\notin U_1\end{aligned}$$

Similarly, assume

$$\begin{aligned}\vec{u}_1 + \vec{u}_2 &\in U_2 \\ \Rightarrow (\vec{u}_1 + \vec{u}_2) - \vec{u}_2 &\in U_2 \\ \Rightarrow \vec{u}_1 &\in U_2 \\ \Rightarrow a \text{ contradiction} \\ \Rightarrow \vec{u}_1 + \vec{u}_2 &\notin U_2\end{aligned}$$

Hence, $U_1 \cup U_2$ is not a subspace of V .

Corollary

Let W_1, W_2 be subspaces of a vector space V and suppose that $W_1 \cup W_2$ is also a subspace of V .

Then, either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Linear combination

Definition

Let $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n \in \mathbb{R}^n$ and $\vec{u} = \lambda_1 \vec{u}_1 + \lambda_2 \vec{u}_2 + \dots + \lambda_n \vec{u}_n$ where $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}$.

Then, we say that $\vec{u}$ is a linear combination of $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$.

Examples

$$1. \quad (1, 3, 4) = 2(3, 2, 1) - 5\left(1, \frac{1}{5}, -\frac{2}{5}\right)$$

So, $(1, 3, 4)$ is a linear combination of $(3, 2, 1)$ and $\left(1, \frac{1}{5}, -\frac{2}{5}\right)$

$$2. \quad \text{Is } (1, 0, 1) \text{ a linear combination of } (3, 2, 1) \text{ and } \left(1, \frac{1}{5}, -\frac{2}{5}\right)?$$

Solution

Let

$$\begin{aligned} (1, 0, 1) &= \lambda_1 (3, 2, 1) + \lambda_2 \left(1, \frac{1}{5}, -\frac{2}{5}\right) \\ \Rightarrow (1, 0, 1) &= (3\lambda_1, 2\lambda_1, \lambda_1) + \left(\lambda_2, \frac{\lambda_2}{5}, -\frac{2\lambda_2}{5}\right) \\ &= \left(3\lambda_1 + \lambda_2, 2\lambda_1 + \frac{\lambda_2}{5}, \lambda_1 - \frac{2\lambda_2}{5}\right) \\ \Rightarrow 1 &= 3\lambda_1 + \lambda_2 \dots\dots\dots (i) \\ 0 &= 2\lambda_1 + \frac{\lambda_2}{5} \dots\dots\dots (ii) \\ 1 &= \lambda_1 - \frac{2\lambda_2}{5} \dots\dots\dots (iii) \end{aligned}$$

From

$$\begin{aligned} (ii) \Rightarrow \frac{\lambda_2}{5} &= -2\lambda_1 \\ \Rightarrow \lambda_2 &= -10\lambda_1 \end{aligned}$$

Substituting this into (i) gives

$$\begin{aligned} 1 &= 3\lambda_1 - 10\lambda_1 \\ \Rightarrow -7\lambda_1 &= 1 \\ \Rightarrow \lambda_1 &= -\frac{1}{7} \end{aligned}$$

From

$$\lambda_2 = -10\lambda_1 \text{ and } \lambda_1 = -\frac{1}{7}$$

$$\Rightarrow \lambda_2 = -10\left(-\frac{1}{7}\right) = \frac{10}{7}$$

Substituting for λ_1 and λ_2 in (iii) we get

$$1 = -\frac{1}{7} - \frac{2}{5}\left(\frac{10}{7}\right)$$

$$= -\frac{1}{7} - \frac{4}{7}$$

$$= -\frac{5}{7}$$

But $1 \neq -\frac{5}{7}$.

Therefore, $(1, 0, 1)$ is not a linear combination of $(3, 2, 1)$ and $\left(1, \frac{1}{5}, -\frac{2}{5}\right)$.

Linear dependence

Definition

Let $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n \in \mathbb{R}^n$. We say that $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ are linearly dependent if one of the vectors is a linear combination of the others.

Theorem

$\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are linearly dependent if and only if there exists some constants $\lambda_1, \lambda_2, \dots, \lambda_n$ not all zero, such that $\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \dots + \lambda_n \vec{v}_n = \vec{0}$.

Proof

Assume $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are linearly dependent say,

$$\vec{v}_n = \lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \dots + \lambda_{n-1} \vec{v}_{n-1}$$

$$\Rightarrow \vec{0} = \lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \dots + \lambda_{n-1} \vec{v}_{n-1} - \vec{v}_n$$

$$\Rightarrow \vec{0} = \lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \dots + \lambda_{n-1} \vec{v}_{n-1} + \lambda_n \vec{v}_n$$

where $\lambda_n = -1 \neq 0$.

Hence, not all λ 's are zero.

Conversely, assume that $\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \dots + \lambda_n \vec{v}_n = \vec{0}$ where not all λ 's are zero, say $\lambda_n \neq 0$.

Then,

$$\begin{aligned}\lambda_n \vec{v}_n &= -\lambda_1 \vec{v}_1 - \lambda_2 \vec{v}_2 - \dots - \lambda_{n-1} \vec{v}_{n-1} \\ \Rightarrow \vec{v}_n &= -\frac{\lambda_1}{\lambda_n} \vec{v}_1 - \frac{\lambda_2}{\lambda_n} \vec{v}_2 - \dots - \frac{\lambda_{n-1}}{\lambda_n} \vec{v}_{n-1}.\end{aligned}$$

This shows that $\vec{v}_n$ is a linear combination of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_{n-1}$ and so $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are linearly dependent.

Remark

1. $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are linearly dependent if one of the vectors is a linear combination of the others.
2. $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are linearly dependent if we can solve the equation $\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \dots + \lambda_n \vec{v}_n = \vec{0}$ and find that not all λ 's are zero.

Linear independence

Definition

$\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m$ are linearly independent if they are not linearly dependent. Thus, $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m$ are linearly independent if for

$$\begin{aligned}\lambda_1 \vec{u}_1 + \lambda_2 \vec{u}_2 + \dots + \lambda_m \vec{u}_m &= \vec{0} \\ \Rightarrow \lambda_1 = \lambda_2 = \dots = \lambda_m &= 0\end{aligned}$$

Examples

1. Check whether $(1,1,0), (1,0,1), (0,1,1)$ are linearly dependent or not

Solution

Let

$$\begin{aligned}\alpha_1 (1,1,0) + \alpha_2 (1,0,1) + \alpha_3 (0,1,1) &= (0,0,0) \\ \Rightarrow (\alpha_1, \alpha_1, 0) + (\alpha_2, 0, \alpha_2) + (0, \alpha_3, \alpha_3) &= (0,0,0) \\ \Rightarrow \alpha_1 + \alpha_2 = 0 \quad \Rightarrow \alpha_1 = -\alpha_2 \dots\dots\dots (i) \\ \alpha_1 + \alpha_3 = 0 \quad \Rightarrow \alpha_1 = -\alpha_3 \dots\dots\dots (ii) \\ \alpha_2 + \alpha_3 = 0 \quad \Rightarrow \alpha_2 = -\alpha_3 \dots\dots\dots (iii)\end{aligned}$$

Taking equations (i) and (ii) we get $\alpha_2 = \alpha_3$, contracting equation (iii).

Hence, $\alpha_1 = \alpha_2 = \alpha_3 = 0$ and thus the vectors are linearly independent.

2. Check whether the vectors $(3,1,2,4)$, $(1,1,1,1)$, $(4,0,2,6)$, $(1,2,1,0)$ are linearly dependent or not

Solution

Let

$$\lambda_1 (3,1,2,4) + \lambda_2 (1,1,1,1) + \lambda_3 (4,0,2,6) + \lambda_4 (1,2,1,0) = (0,0,0,0)$$

$$\Rightarrow 3\lambda_1 + \lambda_2 + 4\lambda_3 + \lambda_4 = 0 \dots\dots(i)$$

$$\lambda_1 + \lambda_2 + 2\lambda_4 = 0 \dots\dots(ii)$$

$$2\lambda_1 + \lambda_2 + 2\lambda_3 + \lambda_4 = 0 \dots\dots(iii)$$

$$4\lambda_1 + \lambda_2 + 6\lambda_3 = 0 \dots\dots(iv)$$

Subtracting equation (iii) from equation (i) gives

$$\lambda_1 + 2\lambda_3 = 0 \Rightarrow \lambda_1 = -2\lambda_3 \dots\dots(v)$$

Subtracting equation (iv) from equation (ii) we get

$$3\lambda_1 + 6\lambda_3 - 2\lambda_4 = 0 \dots\dots(vi)$$

From (v) and (vi) we obtain

$$-3(2\lambda_3) + 6\lambda_3 - 2\lambda_4 = 0$$

$$\Rightarrow -2\lambda_4 = 0$$

$$\Rightarrow \lambda_4 = 0$$

Using equation (ii) and $\lambda_4 = 0$ we have

$$\lambda_1 = -\lambda_2$$

We know that

$$\lambda_1 = -2\lambda_3.$$

So by putting

$$\lambda_3 = 1 \Rightarrow \lambda_1 = -2, \lambda_2 = 2$$

Thus,

$$\lambda_1 = -2, \lambda_2 = 2, \lambda_3 = 1 \text{ and } \lambda_4 = 0$$

and hence the vectors are linearly dependent.

Facts

The following are some facts on linear dependence and independence of vectors;

1. $\{\vec{0}, \vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is a linearly dependent since the set of vectors is having a zero vector.

For if $\lambda_0 \vec{0} + \lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \dots + \lambda_n \vec{v}_n = \vec{0}$, we can put any non-zero value for λ_0 and make the other λ 's zero.

2. $\vec{u}_1, \lambda \vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ are linearly dependent. For if we let $\lambda_0 \vec{u}_1 + \lambda_1 (\lambda \vec{u}_1) + \lambda_2 \vec{u}_2 + \dots + \lambda_n \vec{u}_n = \vec{0}$ and put $\lambda_0 \neq -\lambda_1 \lambda$ and $\lambda_2 = \lambda_3 = \dots = \lambda_n = 0$ then not all λ 's are zero.

3. If $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_r$ are linearly dependent, then $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_r, \vec{u}_{r+1}, \vec{u}_{r+2}, \dots, \vec{u}_n$ are linearly dependent.

Assume $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_r$ are linearly dependent, then $\lambda_1 \vec{u}_1 + \lambda_2 \vec{u}_2 + \dots + \lambda_r \vec{u}_r = \vec{0}$ where not all λ 's are zero.

Thus, in $\lambda_1 \vec{u}_1 + \lambda_2 \vec{u}_2 + \dots + \lambda_r \vec{u}_r + \lambda_{r+1} \vec{u}_{r+1} + \lambda_{r+2} \vec{u}_{r+2} + \dots + \lambda_n \vec{u}_n = \vec{0}$ we can take

$\lambda_{r+1} = \lambda_{r+2} = \dots = \lambda_n = 0$. But $\lambda_1, \lambda_2, \dots, \lambda_r$ are not all zero. Hence,

$\vec{u}_1, \vec{u}_2, \dots, \vec{u}_r, \vec{u}_{r+1}, \vec{u}_{r+2}, \dots, \vec{u}_n$ are linearly dependent.

4. If $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r, \vec{v}_{r+1}, \vec{v}_{r+2}, \dots, \vec{v}_n$ are linearly independent, then $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r$ are linearly independent.

Let

$$\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \dots + \lambda_r \vec{v}_r = \vec{0}.$$

We know that

$$\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \dots + \lambda_r \vec{v}_r + \lambda_{r+1} \vec{v}_{r+1} + \lambda_{r+2} \vec{v}_{r+2} + \dots + \lambda_n \vec{v}_n = \vec{0}$$

where all λ 's are zero and so $\lambda_1 = \lambda_2 = \dots = \lambda_r = 0$.

Hence, $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r$ are linearly independent.

Some results on linear independence

Theorem

Suppose $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ are non-zero vectors such that $\vec{u}_i \cdot \vec{u}_j = 0$ whenever $i \neq j$, then $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ are linearly independent.

Proof

Assume $\lambda_1 \vec{u}_1 + \lambda_2 \vec{u}_2 + \dots + \lambda_n \vec{u}_n = \vec{0}$, then

$$\begin{aligned}\vec{u}_1 \cdot (\lambda_1 \vec{u}_1 + \lambda_2 \vec{u}_2 + \dots + \lambda_n \vec{u}_n) &= 0 \\ \Rightarrow \lambda_1 \vec{u}_1 \cdot \vec{u}_1 + \lambda_2 \vec{u}_1 \cdot \vec{u}_2 + \dots + \lambda_n \vec{u}_1 \cdot \vec{u}_n &= 0 \\ \Rightarrow \lambda_1 \vec{u}_1 \cdot \vec{u}_1 &= 0 \\ \Rightarrow \lambda_1 &= 0\end{aligned}$$

Similarly, $\lambda_2 = \lambda_3 = \dots = \lambda_n = 0$ and therefore $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ are linearly independent.

Theorem

Suppose $\vec{u}$ is a linear combination of $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$. Then $\vec{u}$ can be expressed as a linear combination of $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ in only one way if and only if $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ are linearly independent.

Proof

$\Rightarrow$: assume $\vec{u}$ can be expressed/written as a linear combination of the $\vec{u}_i$'s in more than one way. What we want to show is that the $\vec{u}_i$'s are linearly dependent. So we have

$$\begin{aligned}\lambda_1 \vec{u}_1 + \lambda_2 \vec{u}_2 + \dots + \lambda_n \vec{u}_n &= \vec{u} \\ \text{and} \\ \alpha_1 \vec{u}_1 + \alpha_2 \vec{u}_2 + \dots + \alpha_n \vec{u}_n &= \vec{u}\end{aligned}$$

Subtracting, we have

$$(\lambda_1 - \alpha_1) \vec{u}_1 + (\lambda_2 - \alpha_2) \vec{u}_2 + \dots + (\lambda_n - \alpha_n) \vec{u}_n = \vec{0}$$

Since $\lambda_i \neq \alpha_i$ for some i 's, then $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ are linearly dependent.

$\Leftarrow$: suppose $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ are linearly independent and let

$$\begin{aligned}\vec{u} &= \lambda_1 \vec{u}_1 + \lambda_2 \vec{u}_2 + \dots + \lambda_n \vec{u}_n \\ \text{and} \\ \vec{u} &= \alpha_1 \vec{u}_1 + \alpha_2 \vec{u}_2 + \dots + \alpha_n \vec{u}_n\end{aligned}$$

Subtracting, we get

$$\vec{0} = (\lambda_1 - \alpha_1)\vec{u}_1 + (\lambda_2 - \alpha_2)\vec{u}_2 + \dots + (\lambda_n - \alpha_n)\vec{u}_n$$

and so

$$\begin{aligned} \lambda_1 - \alpha_1 = 0, \lambda_2 - \alpha_2 = 0, \dots, \lambda_n - \alpha_n = 0 \\ \Rightarrow \lambda_1 = \alpha_1, \lambda_2 = \alpha_2, \dots, \lambda_n = \alpha_n \end{aligned}$$

Thus, $\vec{u}$ can be expressed as a linear combination of $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ in only one way.

Theorem

Let $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n \in \mathbb{R}^n$. Let A be the set of all linear combinations of $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$. Then, A is a subspace of $\mathbb{R}^n$.

Proof

Let

$$\begin{aligned} \vec{x}, \vec{y} &\in A \\ \Rightarrow \vec{x} &= \lambda_1 \vec{u}_1 + \lambda_2 \vec{u}_2 + \dots + \lambda_n \vec{u}_n \\ \vec{y} &= \beta_1 \vec{u}_1 + \beta_2 \vec{u}_2 + \dots + \beta_n \vec{u}_n \end{aligned}$$

Now,

$$\vec{x} + \vec{y} = (\lambda_1 + \beta_1)\vec{u}_1 + (\lambda_2 + \beta_2)\vec{u}_2 + \dots + (\lambda_n + \beta_n)\vec{u}_n \in A$$

So, the 1st condition is satisfied.

Next, we have

$$\lambda \vec{x} = (\lambda \lambda_1)\vec{u}_1 + (\lambda \lambda_2)\vec{u}_2 + \dots + (\lambda \lambda_n)\vec{u}_n \in A$$

Thus, the 2nd condition holds.

Hence, A is a vector subspace of $\mathbb{R}^n$.

Note

We say that A is a vector subspace spanned by $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$.

Spanning set

Definition

Let $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ be a finite set of a vector space V . if S is a set of elements of the form $\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \dots + \lambda_n \vec{v}_n$ then we say that S is spanned by $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$.

The set $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is then called a spanning set of S .

Examples

1. $S = \{1, x, x^2\}$ spans all polynomials of degree less than or equal to 2

Note

A spanning set is not necessarily unique. For instance, let $V = \{x-1, x+2, 2x^2+x\}$, then

$$1 = -\frac{1}{3}(x-1) + \frac{1}{3}(x+2) + 0(2x^2+x)$$

$$x = \frac{2}{3}(x-1) + \frac{1}{3}(x+2) + 0(2x^2+x)$$

$$x^2 = -\frac{1}{3}(x-1) - \frac{1}{6}(x+2) + \frac{1}{2}(2x^2+x)$$

Thus, S and V span the same set.

2. $W = \{(1,0,0), (0,1,0), (0,0,1)\}$. W spans $\mathbb{R}^3$, since $(x,y,z) \in \mathbb{R}^3$ can be written as $(x,y,z) = x(1,0,0) + y(0,1,0) + z(0,0,1)$

Note

Let $U = \{(1,0,2), (0,1,0), (1,1,1)\}$. Then

$$(1,0,0) = -(1,0,2) - 2(0,1,0) + 2(1,1,1)$$

$$(0,1,0) = 0(1,0,2) + (0,1,0) + 0(1,1,1)$$

$$(0,0,1) = (1,0,2) + (0,1,0) - (1,1,1)$$

So both W and U span $\mathbb{R}^3$.

Definition

If a set V can be spanned by a finite number of elements, then we say that V is a finite dimensional space. If not then it is of infinite dimension.

Note

The largest number of independent vectors in a vector space equals the minimum number of vectors in a spanning set.

Let V be a vector space with S as an independent set in V and T as a spanning set of V . Then $|S| \leq |T|$.

Basis of a vector space

Definition

Let B be a finite subset of a vector space V . We say that B is a basis for V if:

- (i) B spans V
- (ii) B is an independent set

Examples

1. Let $\vec{e}_1 = (1, 0, 0)$, $\vec{e}_2 = (0, 1, 0)$, $\vec{e}_3 = (0, 0, 1)$. Then $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ is a basis for $\mathbb{R}^3$ since if

(i) $\vec{v} = (\alpha_1, \alpha_2, \alpha_3)$ is an arbitrary element in $\mathbb{R}^3$, then

$(\alpha_1, \alpha_2, \alpha_3) = \alpha_1 \vec{e}_1 + \alpha_2 \vec{e}_2 + \alpha_3 \vec{e}_3$, that is, $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ generates $\mathbb{R}^3$

(ii) $\lambda_1 \vec{e}_1 + \lambda_2 \vec{e}_2 + \lambda_3 \vec{e}_3 = \vec{0}$, then $\lambda_1 = \lambda_2 = \lambda_3 = 0$, that is, $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ is linearly independent.

The set $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ is called the standard basis for $\mathbb{R}^3$.

2. In $\mathbb{R}^2$, the standard basis is $\{(1, 0), (0, 1)\}$

3. $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)\}$ is not a basis for $\mathbb{R}^3$ since the vectors are linearly dependent.

4. Let $\vec{v}_1 = (1, 0, 1)$, $\vec{v}_2 = (2, -1, 1)$, $\vec{v}_3 = (4, 1, 1)$, then $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is a basis for $\mathbb{R}^3$. It can be easily verified that

$$(\alpha_1, \alpha_2, \alpha_3) = \frac{1}{4}((-2\alpha_1 + 2\alpha_2 + 6\alpha_3)\vec{v}_1 + (\alpha_1 - 3\alpha_2 - \alpha_3)\vec{v}_2 + (\alpha_1 + \alpha_2 - \alpha_3)\vec{v}_3), \text{ that is,}$$

$\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ generates $\mathbb{R}^3$.

$\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is a linearly independent set, for if

$\lambda_1\vec{v}_1 + \lambda_2\vec{v}_2 + \lambda_3\vec{v}_3 = \vec{0}$ where $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}^3$, then

$$\lambda_1(1, 0, 1) + \lambda_2(2, -1, 1) + \lambda_3(4, 1, 1) = (0, 0, 0)$$

$$\Rightarrow \lambda_1 + 2\lambda_2 + 4\lambda_3 = 0 \dots\dots\dots(i)$$

$$\Rightarrow -\lambda_2 + \lambda_3 = 0 \Rightarrow \lambda_2 = \lambda_3 \dots\dots\dots(ii)$$

$$\Rightarrow \lambda_1 + \lambda_2 + \lambda_3 = 0 \dots\dots\dots(iii)$$

Subtracting equation (iii) from equation (ii) we get $\lambda_2 + 3\lambda_3 = 0 \Rightarrow \lambda_2 = -3\lambda_3$

From $\lambda_2 = \lambda_3$ and $\lambda_2 = -3\lambda_3 \Rightarrow \lambda_2 = \lambda_3 = 0$.

This implies that $\lambda_1 = 0$.

Thus, $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is a linearly independent set.

Remark

- (i) A basis for a vector space is not unique
 - (ii) If $B \subseteq V$, then B is a basis for the vector space V if every vector in V can be written as a linear combination of members of B in only one way.
5. $\{(1, 2, 1), (3, 2, 1)\}$ is not a basis for $\mathbb{R}^3$ because the two vectors cannot span $\mathbb{R}^3$.
6. Is $\{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ a basis for $\mathbb{R}^3$?

Solution

The set is linearly independent.

Take $(x, y, z) \in \mathbb{R}^3$ and let

$$(x, y, z) = \lambda_1(1, 1, 0) + \lambda_2(1, 0, 1) + \lambda_3(0, 1, 1)$$

$$\Rightarrow x = \lambda_1 + \lambda_2 \dots\dots\dots(i)$$

$$\Rightarrow y = \lambda_1 + \lambda_3 \dots\dots\dots(ii)$$

$$\Rightarrow z = \lambda_2 + \lambda_3 \dots\dots\dots(iii)$$

Subtracting equation (ii) from (i) gives

$$x - y = \lambda_2 - \lambda_3 \dots\dots (iv)$$

Adding equation (iv) to equation (iii) we obtain

$$\begin{aligned} x - y + z &= 2\lambda_2 \\ \Rightarrow \lambda_2 &= \frac{x + y + z}{2} \dots\dots (v) \end{aligned}$$

From equation (i) and equation (v), we have

$$\begin{aligned} x &= \lambda_1 + \frac{x + y + z}{2} \\ \Rightarrow \lambda_1 &= \frac{x + y - z}{2} \dots\dots (vi) \end{aligned}$$

Now, from equation (vi) and equation (ii) we get

$$\begin{aligned} y &= \frac{x + y - z}{2} + \lambda_3 \\ \Rightarrow \lambda_3 &= \frac{y - x + z}{2} \end{aligned}$$

So these vectors form a basis for $\mathbb{R}^3$ since every vector in $\mathbb{R}^3$ can be written in terms of the three vectors in only one way.

7. Find a basis for $V = \{(x, y, z) : y = 2z\}$

Solution

The basis is $\{(1, 2, 1), (0, 2, 1)\}$

8. Find a basis for $U = \{(x, y, z) : y = 2z \text{ and } x = 2y\}$

Solution

The basis is $\{(4, 2, 1)\}$

9. Compute a basis for $\{(x, y, z) : 2x - y + 3z = 0\}$

Solution

The basis is $\{(0, 3, 1), (1, 5, 1)\}$

10. Determine a basis for

$$\begin{aligned} &\{(x, y, z, w) : x + y + z + w = 0 \text{ and } x - y + z - w = 0\} \\ &= \{(x, y, z, w) : 2x + 2z = 0 \text{ and } 2y + 2w = 0\} \end{aligned}$$

Solution

The basis is $\{(0,1,0,-1), (-1,0,1,0)\}$

11. Prove that $(1,-1,0)$ and $(0,1,-1)$ span the plane whose equation is $x + y + z = 0$.

Solution

Take a general vector, say, (x, y, z) on the plane. From $x + y + z = 0 \Rightarrow z = -x - y$

Therefore,

$$\begin{aligned}(x, y, z) &= (x, y, -x - y) = \lambda_1 (1, -1, 0) + \lambda_2 (0, 1, -1) \\ \Rightarrow x &= \lambda_1 \\ y &= -\lambda_1 + \lambda_2 \\ -x - y &= -\lambda_2 \Rightarrow x + y = \lambda_2\end{aligned}$$

Therefore,

$$(x, y, -x - y) = x(1, -1, 0) + (x + y)(0, 1, -1)$$

Hence, the two vectors span the plane. In fact, the two vectors form a basis for the plane.

Dimension of a vector space

Definition

Let V be a vector space and B a basis for V . Then, the dimension of V , denoted by $\dim V$, is the number of vectors in B .

Examples

1. Let $\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$. Then, we have 3 choices for any element of $\mathbb{R}^3$. So $\dim \mathbb{R}^3 = 3$.
2. Let $V = \{(x, y, z) : z = x + y\}$. Then, we have 2 choices for any element of V and so $\dim V = 2$. A basis for V is $\{(1, 1, 2), (0, 1, 1)\}$.
3. Suppose $U = \{(x, y, z) : x = 2y \text{ and } z = 3y\}$, then we have only one choice. Thus, $\dim U = 1$. A basis for U is $\{(2, 1, 3)\}$.
4. If W has a basis $\{\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n\}$, then every member of W is of the form $\lambda_1 \vec{a}_1 + \lambda_2 \vec{a}_2 + \dots + \lambda_n \vec{a}_n$. Here, we have n choices and so $\dim W = n$.

5. Let $Z = \{(x, y, z, w) : x + y + z + w = 0\}$, the $\dim Z = 3$ and a basis for Z is

$$\{(1, 0, 0, -1), (0, 0, -1, 1), (1, -1, 0, 0)\}.$$

6. Suppose

$$\begin{aligned} A &= \{(x, y, z, w, t) : x + y + z + w + t = 0 \text{ and } x - y + z - w + t = 0\} \\ &= \{(x, y, z, w, t) : 2x + 2z + 2t = 0 \text{ and } 2y + 2w = 0\}. \end{aligned}$$

Hence, $\dim A = 3$. A basis for A is $\{(0, 1, 0, -1, 0), (0, 0, 1, 0, -1), (1, 0, 0, 0, -1)\}$.

Definition

Let U, V be subspaces of $\mathbb{R}^n$. Then, the sum $U + V = \{\vec{u} + \vec{v} : \vec{u} \in U \text{ and } \vec{v} \in V\}$

Theorem

$$\dim(U + V) = \dim U + \dim V - \dim(U \cap V)$$

Proof

Let $\{\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n\}$ be a basis for $U \cap V$, that is, $\dim(U \cap V) = n$.

Extend this to a basis for U by adding $\vec{b}_1, \vec{b}_2, \dots, \vec{b}_r$, that is, $\dim U = n + r$.

Extend this to a basis for V by adding $\vec{c}_1, \vec{c}_2, \dots, \vec{c}_s$, that is, $\dim V = n + s$.

Claim

$\{\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n, \vec{b}_1, \vec{b}_2, \dots, \vec{b}_r, \vec{c}_1, \vec{c}_2, \dots, \vec{c}_s\}$ form a basis for $U + V$, that is, $\dim(U + V) = n + r + s$

Proof

To show that the set spans $U + V$, we take $\vec{u} + \vec{v} \in U + V$, where $\vec{u} \in U$ and $\vec{v} \in V$.

We can write

$$\vec{u} = \lambda_1 \vec{a}_1 + \lambda_2 \vec{a}_2 + \dots + \lambda_n \vec{a}_n + \lambda_{n+1} \vec{b}_1 + \lambda_{n+2} \vec{b}_2 + \dots + \lambda_{n+r} \vec{b}_r$$

and

$$\vec{v} = \alpha_1 \vec{a}_1 + \alpha_2 \vec{a}_2 + \dots + \alpha_n \vec{a}_n + \alpha_{n+1} \vec{c}_1 + \alpha_{n+2} \vec{c}_2 + \dots + \alpha_{n+s} \vec{c}_s.$$

Then,

$$\begin{aligned} \vec{u} + \vec{v} = & (\lambda_1 + \alpha_1) \vec{a}_1 + (\lambda_2 + \alpha_2) \vec{a}_2 + \dots + (\lambda_n + \alpha_n) \vec{a}_n \\ & + \lambda_{n+1} \vec{b}_1 + \lambda_{n+2} \vec{b}_2 + \dots + \lambda_{n+r} \vec{b}_r + \alpha_{n+1} \vec{c}_1 + \alpha_{n+2} \vec{c}_2 + \dots + \alpha_{n+s} \vec{c}_s. \end{aligned}$$

Hence, every element in $U+V$ can be written as a linear combination of $\vec{a}_i$'s, $\vec{b}_j$'s and $\vec{c}_k$'s.

To prove linear independence, let

$$\vec{a} = \lambda_1 \vec{a}_1 + \lambda_2 \vec{a}_2 + \dots + \lambda_n \vec{a}_n, \vec{b} = \alpha_1 \vec{b}_1 + \alpha_2 \vec{b}_2 + \dots + \alpha_r \vec{b}_r \text{ and } \vec{c} = \tau_1 \vec{c}_1 + \tau_2 \vec{c}_2 + \dots + \tau_s \vec{c}_s.$$

Then, $\lambda_1 \vec{a}_1 + \lambda_2 \vec{a}_2 + \dots + \lambda_n \vec{a}_n + \alpha_1 \vec{b}_1 + \alpha_2 \vec{b}_2 + \dots + \alpha_r \vec{b}_r + \tau_1 \vec{c}_1 + \tau_2 \vec{c}_2 + \dots + \tau_s \vec{c}_s = \vec{0}$ becomes

$$\vec{a} + \vec{b} + \vec{c} = \vec{0}.$$

We know $\vec{b} \in U$. Then, from $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, we have

$$\begin{aligned} \vec{b} &= -\vec{a} - \vec{c} \\ \Rightarrow \vec{b} &\in \vec{b} \\ \Rightarrow \vec{b} &\in U \cap V. \end{aligned}$$

So we can write

$$\begin{aligned} \vec{b} &= \rho_1 \vec{a}_1 + \rho_2 \vec{a}_2 + \dots + \rho_n \vec{a}_n \\ \Rightarrow \alpha_1 \vec{b}_1 + \alpha_2 \vec{b}_2 + \dots + \alpha_r \vec{b}_r - \rho_1 \vec{a}_1 + \rho_2 \vec{a}_2 + \dots + \rho_n \vec{a}_n &= \vec{0} \end{aligned}$$

But $\vec{b}_1, \vec{b}_2, \dots, \vec{b}_r, \vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are linearly independent. So, $\alpha_1 = \alpha_2 = \dots = \alpha_r = 0$.

Similarly, $\tau_1 = \tau_2 = \dots = \tau_s = 0$.

So $\lambda_1 \vec{a}_1 + \lambda_2 \vec{a}_2 + \dots + \lambda_n \vec{a}_n = \vec{0}$. But $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are linearly independent. This implies

$$\lambda_1 = \lambda_2 = \dots = \lambda_n = 0.$$

Hence, the $\vec{a}_i$'s, $\vec{b}_j$'s and $\vec{c}_k$'s are linearly independent.

Thus, the claim is true.

Therefore,

$$\begin{aligned}\dim(U + V) &= n + r + s \\ &= (n + r) + (n + s) - n \\ &= \dim U + \dim V - \dim(U \cap V)\end{aligned}$$

Example

Let

$$\begin{aligned}U &= \{(x, y, z) : x + y - z = 0\} \\ V &= \{(x, y, z) : 3x - y - z = 0\}\end{aligned}$$

Determine bases for $U \cap V$, U , V , $U + V$.

Solution

$$\begin{aligned}U \cap V &= \{(x, y, z) : x + y - z = 0 \text{ and } 3x - y - z = 0\} \\ &= \{(x, y, z) : 4x - 2z = 0 \text{ and } 2x - 2y = 0\}\end{aligned}$$

Bases for $U \cap V$ is $\{(1, 1, 2)\}$

Bases for U is $\{(1, 1, 2), (1, 0, 1)\}$

Bases for V is $\{(1, 1, 2), (0, 1, -1)\}$

Bases for $U + V$ is $\{(1, 1, 2), (1, 0, 1), (0, 1, -1)\}$

Matrices

Definition

A matrix is an array of vectors written vertically.

Matrix entries

Let A be a $m \times n$ matrix, that is, a matrix with m rows and n columns. Then the scalar entry in the i th row and j th column of A is denoted by a_{ij} and is called the (i, j) -entry of A .

$$A = \begin{pmatrix} a_{11} & \cdot & \cdot & a_{1j} & \cdot & \cdot & a_{1n} \\ \cdot & & & & \cdot & \cdot & \cdot \\ a_{i1} & & & a_{ij} & \cdot & \cdot & a_{in} \\ \cdot & & & & & & \cdot \\ a_{m1} & \cdot & \cdot & a_{mj} & \cdot & \cdot & a_{mn} \end{pmatrix} = (a_{ij})$$

For instance, the (3,2)-entry is the number a_{32} in the third row, second column. The columns of A are vectors in $\mathbb{R}^m$ and are denoted by $\vec{a}_1, \dots, \vec{a}_n$ we can write $A = (\vec{a}_1, \dots, \vec{a}_n)$.

Note

The number a_{ij} is the i th entry (from the top) of the j th column of vector $\vec{a}_j$.

The diagonal entries in a $m \times n$ matrix $A = (a_{ij})$ are $a_{11}, a_{22}, a_{33}, \dots$ and they form the main diagonal of A . A diagonal matrix is a square matrix whose non-diagonal entries are zero

A $m \times n$ matrix whose entries are all zero is a zero matrix and is denoted by 0 .

Sums and scalar multiples

Let A, B and C be matrices of the same size/order and let r and s be scalars. Then

- (i) $A+B=B+A$
- (ii) $(A+B)+C=A+(B+C)$
- (iii) $A+0=A$
- (iv) $r(A+B)=rA+rB$
- (v) $(r+s)A=rA+sA$
- (vi) $r(sA)=(rs)A$

Example

Let $A = \begin{pmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{pmatrix}$ and $C = \begin{pmatrix} 2 & -3 \\ 0 & 1 \end{pmatrix}$. Compute

- (i) $A+B$
- (ii) $A+C$
- (iii) $2B$

(iv) $A-2B$

Solution

(i) The value of $A+B$ is given by

$$\begin{aligned} A+B &= \begin{pmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{pmatrix} \\ &= \begin{pmatrix} 5 & 1 & 6 \\ 2 & 8 & 9 \end{pmatrix} \end{aligned}$$

(ii) $A+C$ is not defined because A and C have different sizes

(iii) The value of $2B$ is given by

$$\begin{aligned} 2B &= 2 \begin{pmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 2 & 2 \\ 6 & 10 & 14 \end{pmatrix} \end{aligned}$$

(iv) The value of $A-2B$ is given by

$$\begin{aligned} A-2B &= \begin{pmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{pmatrix} - 2 \begin{pmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 2 & 2 \\ 6 & 10 & 14 \end{pmatrix} \\ &= \begin{pmatrix} 2 & -2 & 3 \\ -7 & -7 & -12 \end{pmatrix} \end{aligned}$$

Matrix multiplication

Example

(i) Compute AB , where $A = \begin{pmatrix} 2 & 3 \\ 1 & -5 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{pmatrix}$

(ii) Let A be a 3×5 matrix and B be a 5×2 matrix. What are the sizes of AB and BA , if they are defined?

Solution

(i) Let $A = \begin{pmatrix} 2 & 3 \\ 1 & -5 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{pmatrix}$. Then, $AB = A(\vec{b}_1 \vec{b}_2 \vec{b}_3)$ where

$$\vec{b}_1 = \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \vec{b}_2 = \begin{pmatrix} 3 \\ -2 \end{pmatrix} \text{ and } \vec{b}_3 = \begin{pmatrix} 6 \\ 3 \end{pmatrix}. \text{ Thus, } AB = A(\vec{b}_1 \vec{b}_2 \vec{b}_3) = \begin{pmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \end{pmatrix}$$

(ii) Since A has 5 columns and B has 5 rows, the product AB is defined and is a 3×2 matrix.

The product BA is not defined since the 2 columns of B do not match the 3 rows of A.

Properties of matrix multiplication

Let A be a $m \times n$ matrix and let B and C have sizes for which the indicated sums and products are defined. Then

- (i) $A(BC) = (AB)C$ (Associative law of multiplication)
- (ii) $A(B+C) = AB+AC$ (Left distributive law)
- (iii) $(B+C)A = BA+CA$ (Right distributive law)
- (iv) $r(AB) = (rA)B = A(rB)$ for any scalar r
- (v) $I_n A = A = A I_n$ (Identity for matrix multiplication)

Example

Let $A = \begin{pmatrix} 5 & 1 \\ 3 & -2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 4 & 3 \end{pmatrix}$. Show that these matrices do not commute, that is,

verify $AB \neq BA$

Solution

Let $A = \begin{pmatrix} 5 & 1 \\ 3 & -2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 4 & 3 \end{pmatrix}$. Then $AB = \begin{pmatrix} 5 & 1 \\ 3 & -2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 14 & 3 \\ -2 & -6 \end{pmatrix}$ and

$$BA = \begin{pmatrix} 2 & 0 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} 5 & 1 \\ 3 & -2 \end{pmatrix} = \begin{pmatrix} 10 & 2 \\ 29 & -2 \end{pmatrix}.$$

This shows that $AB \neq BA$.

Note

If (i) $AB = AC$ then it is not true in general that $B = C$

(ii) $AB = 0$, we cannot conclude in general that either $A = 0$ or $B = 0$.

Powers of a matrix

Let A be a $m \times n$ matrix and k a positive integer. Then, A^k denotes the product of k copies of A .

Transpose of a matrix

Given a $m \times n$ matrix A , the transpose of A is the $n \times m$ matrix, denoted by A^T , whose columns are formed from corresponding rows of A .

Example

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $B = \begin{pmatrix} -5 & 2 \\ 1 & -3 \\ 0 & 4 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -3 & 5 & -2 & 7 \end{pmatrix}$. Find the transpose of A , B and C

Solution

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $B = \begin{pmatrix} -5 & 2 \\ 1 & -3 \\ 0 & 4 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -3 & 5 & -2 & 7 \end{pmatrix}$. The

$$A^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}, B^T = \begin{pmatrix} -5 & 1 & 0 \\ 2 & -3 & 4 \end{pmatrix} \text{ and } C^T = \begin{pmatrix} 1 & -3 \\ 1 & 5 \\ 1 & -2 \\ 1 & 7 \end{pmatrix}$$

Theorem

Let A and B denote matrices whose orders are appropriate for the following sums and products, then

$$(i) \quad (A^T)^T = A$$

$$(ii) \quad (A + B)^T = A^T + B^T$$

$$(iii) \quad \text{For any scalar } r, (rA)^T = rA^T$$

$$(iv) \quad (AB)^T = B^T A^T$$

Determinants

Recall that a 2x2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is invertible if and only if $ad - bc \neq 0$ and the expression $ad - bc$ is called the determinant of matrix A. This determinant is denoted by

$$\det(A) = ad - bc \quad \text{or} \quad \left| \begin{pmatrix} a & b \\ c & d \end{pmatrix} \right| = ad - bc$$

And the inverse of A can be expressed in terms of the determinant as $A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

Example

Compute the determinant and inverse of $A = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}$

Solution

The determinant of $A = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}$ is $\det A = 3(6) - 4(5) = -2 \neq 0$, and so A is invertible.

Therefore,

$$\begin{aligned} A^{-1} &= \frac{1}{-2} \begin{pmatrix} 6 & -4 \\ -5 & 3 \end{pmatrix} \\ &= \begin{pmatrix} -3 & 2 \\ 5/2 & -3/2 \end{pmatrix} \end{aligned}$$

Minors and Cofactors

Definition

If A is a square matrix, then the minor of entry a_{ij} is denoted by M_{ij} and defined to be the determinant of the submatrix that remains after the i th row and j th column are deleted from A.

The number $(-1)^{i+j} M_{ij}$ is denoted by C_{ij} and is called the cofactor of entry a_{ij}

Example

Let $A = \begin{pmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{pmatrix}$, then the minor of entry a_{11} is $M_{11} = \begin{vmatrix} 5 & 6 \\ 4 & 8 \end{vmatrix} = 16$

The cofactor of a_{11} is $C_{11} = (-1)^{1+1} M_{11} = M_{11} = 16$

Similarly, the minor of entry a_{32} is $M_{32} = \begin{vmatrix} 3 & -4 \\ 2 & 6 \end{vmatrix} = 26$

The cofactor of a_{32} is $C_{32} = (-1)^{3+2} M_{32} = -M_{32} = -26$

General determinant

Definition

If A is an $n \times n$ matrix, then the number obtained by multiplying the entries in any row or column of A by the corresponding cofactors and adding the resulting products is called the determinant of A, and the sums themselves are called cofactor expansions of A. That is,

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \dots + a_{nj}C_{nj} \text{ (cofactor expansion along the } j\text{th column)}$$

And

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in} \text{ (cofactor expansion along the } i\text{th row)}$$

Example

Determine the determinant of matrix $A = \begin{pmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{pmatrix}$ by cofactor expansion along the first row.

Solution

$$\begin{aligned} \det(A) &= \begin{vmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{vmatrix} = 3 \begin{vmatrix} -4 & 3 \\ 4 & -2 \end{vmatrix} - 1 \begin{vmatrix} -2 & 3 \\ 5 & -2 \end{vmatrix} + 0 \begin{vmatrix} -2 & -4 \\ 5 & 4 \end{vmatrix} \\ &= 3(-4) - 1(-11) + 0 \\ &= -1 \end{aligned}$$

Note

If A is an $n \times n$ triangular matrix (upper triangular, lower triangular or diagonal), then $\det(A)$ is the product of the entries on the main diagonal of the matrix. That is, $\det(A) = a_{11}a_{22}\dots a_{nn}$

Evaluating determinants by row reduction

In this method we evaluate a determinant by reducing the associated matrix to row echelon form.

Elementary row operations

Let A be an $n \times n$ matrix. If

- (a) B is the matrix that results when a single row or single column of A is multiplied by a scalar k , then $\det(B) = k \det(A)$
- (b) B is the matrix that results when two rows or two columns of A are interchanged, then $\det(B) = -\det(A)$
- (c) B is the matrix that results when a multiple of one row of A is added to another row or when a multiple of one column is added to another column, then $\det(B) = \det(A)$
- (d) A has a row or column of zero, then $\det(A) = 0$
- (e) A has identical rows or columns, then $\det(A) = 0$

Example

Evaluate $\det(A)$ where $A = \begin{pmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{pmatrix}$

Solution

We will reduce A to row echelon form (which is upper triangular). Therefore,

$$\begin{aligned} \det(A) &= \begin{vmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{vmatrix} \\ &= - \begin{vmatrix} 3 & -6 & 9 \\ 0 & 1 & 5 \\ 5 & 4 & -2 \end{vmatrix} \quad \text{the first and second rows of } A \text{ are interchanged} \\ &= -3 \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 2 & 6 & 1 \end{vmatrix} \quad \text{a common factor of 3 from first row is taken out of the determinant sign} \\ &= -3 \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 10 & -5 \end{vmatrix} \quad \text{-2 times the first row is added to the third row} \\ &= -3 \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & -55 \end{vmatrix} \quad \text{-10 times the second row is added to the third row} \\ &= (-3)(1)(1)(-55) \\ &= 165 \end{aligned}$$

Note

If A and B are n square matrices, then $\det(AB) = \det(A)\det(B)$

Adjoint of a matrix

If A is an $n \times n$ matrix and C_{ij} is the cofactor of a_{ij} , then the matrix

$$\begin{pmatrix} C_{11} & C_{12} & \cdot & \cdot & \cdot & C_{1n} \\ C_{21} & C_{22} & \cdot & \cdot & \cdot & C_{2n} \\ \cdot & \cdot & & & & \cdot \\ \cdot & \cdot & & & & \cdot \\ \cdot & \cdot & & & & \cdot \\ C_{n1} & C_{n2} & \cdot & \cdot & \cdot & C_{nn} \end{pmatrix}$$

Is called the matrix of cofactors of A.

The transpose of this matrix is called the Adjoint (or Classical Adjoint) of A and is denoted by Adj(A).

Example

Compute the adjoint of $A = \begin{pmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{pmatrix}$

Solution

The cofactors of $A = \begin{pmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{pmatrix}$ are

$$C_{11} = 12 \quad C_{12} = 6 \quad C_{13} = -16$$

$$C_{21} = 4 \quad C_{22} = 2 \quad C_{23} = 16$$

$$C_{31} = 12 \quad C_{32} = -10 \quad C_{33} = 16$$

So the matrix of cofactors is $\begin{pmatrix} 12 & 6 & -16 \\ 4 & 2 & 16 \\ 12 & -10 & 16 \end{pmatrix}$ and the adjoint of A is

$$\text{Adj}(A) = \begin{pmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{pmatrix}$$

Inverse of a matrix using its adjoint

If A is an invertible matrix, then $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$

Example

Compute the inverse of the matrix in the example above.

Solution

$$A = \begin{pmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{pmatrix} \text{ and the cofactors are } \begin{matrix} C_{11} = 12 & C_{12} = 6 & C_{13} = -16 \\ C_{21} = 4 & C_{22} = 2 & C_{23} = 16 \\ C_{31} = 12 & C_{32} = -10 & C_{33} = 16 \end{matrix}$$

Thus, determinant of A ,

$$\begin{aligned} \det(A) &= (3 \times 12) + (2 \times 6) + (-1 \times -16) \\ &= 36 + 12 + 16 \\ &= 64 \end{aligned}$$

And

$$\text{Adj}(A) = \begin{pmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{pmatrix}$$

Hence,

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = \frac{1}{64} \begin{pmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{pmatrix} = \begin{pmatrix} \frac{12}{64} & \frac{4}{64} & \frac{12}{64} \\ \frac{6}{64} & \frac{2}{64} & \frac{-10}{64} \\ \frac{-16}{64} & \frac{16}{64} & \frac{16}{64} \end{pmatrix}$$

Theorem

(a) Let A be an invertible matrix. Then A^{-1} is invertible and $(A^{-1})^{-1} = A$

(b) If A and B are $n \times n$ invertible matrices, then so is AB and the inverse of AB is the product of the inverses of A and B in the reverse order, that is, $(AB)^{-1} = B^{-1}A^{-1}$

(c) Let A be an invertible matrix. Then, the transpose of A , that is A^T , is also invertible and its inverse is the transpose of A^{-1} , that is, $(A^T)^{-1} = (A^{-1})^T$

Definition

The trace of an n -square matrix A , denoted by $\text{tr}(A)$, is the sum of its diagonal elements.

Solving equations

Theorem

Let A be an invertible $n \times n$ matrix. Then for each $\vec{b}$ in $\mathbb{R}^n$, the equation $A\vec{x} = \vec{b}$ has the unique solution $\vec{x} = A^{-1}\vec{b}$.

Example

Solve the system

$$3x_1 + 4x_2 = 3$$

$$5x_1 + 6x_2 = 7$$

Solution

This system is equivalent to $A\vec{x} = \vec{b}$ where $A = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}$, $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$.

In this case $A^{-1} = \begin{pmatrix} -3 & 2 \\ 5/2 & -3/2 \end{pmatrix}$ and so

$$\begin{aligned} \vec{x} &= A^{-1}\vec{b} \\ &= \begin{pmatrix} -3 & 2 \\ 5/2 & -3/2 \end{pmatrix} \begin{pmatrix} 3 \\ 7 \end{pmatrix} \\ &= \begin{pmatrix} 5 \\ -3 \end{pmatrix}. \end{aligned}$$

Gauss elimination method

Given a system of equations

$$a_1x + b_1y + c_1z = d_1 \dots\dots\dots (i)$$

$$a_2x + b_2y + c_2z = d_2 \dots\dots\dots (ii)$$

$$a_3x + b_3y + c_3z = d_3 \dots\dots\dots (iii)$$

We wish to solve for the unknowns x, y and z.

To do so, we can multiply equation (iii) by $\frac{c_2}{c_3}$ then subtract it from equation (ii) to get

$$a_1x + b_1y + c_1z = d_1 \dots\dots\dots (i)$$

$$a_2x + b_2y + c_2z = d_2 \dots\dots\dots (ii)$$

$$a'_3x + b'_3y = d'_3 \dots\dots\dots (iii)'$$

Next, we multiply equation (ii) by $\frac{c_1}{c_2}$ and subtract it from equation (i) to obtain

$$a_1x + b_1y + c_1z = d_1 \dots\dots\dots (i)$$

$$a'_2x + b'_2y = d'_2 \dots\dots\dots (ii)'$$

$$a'_3x + b'_3y = d'_3 \dots\dots\dots (iii)'$$

Finally, we multiply equation (iii)' by $\frac{b'_2}{b'_3}$ and subtract it from equation (ii)' to have

$$a_1x + b_1y + c_1z = d_1 \dots\dots\dots (i)$$

$$a'_2x + b'_2y = d'_2 \dots\dots\dots (ii)'$$

$$a''_3x = d''_3 \dots\dots\dots (iii)''$$

Then, we solve for x, then for y and finally for z.

Example

Solve the following equations for x, y and z

$$\begin{aligned}x - y + 2z &= 1 \\2x + y + z &= 3 \\3x + y + z &= 4\end{aligned}$$

Solution

$$\begin{aligned}x - y + 2z &= 1 \dots\dots (i) \\2x + y + z &= 3 \dots\dots (ii) \\3x + y + z &= 4 \dots\dots (iii)\end{aligned}$$

Subtracting equation (iii) from equation (ii), we get

$$\begin{aligned}x - y + 2z &= 1 \dots\dots (i) \\2x + y + z &= 3 \dots\dots (ii) \\x &= 1 \dots\dots (iv)\end{aligned}$$

Next, subtracting $2 \times \text{equation (ii)}$ from equation (i), we obtain

$$\begin{aligned}x - y + 2z &= 1 \dots\dots (i) \\-3x - 3y &= -5 \dots\dots (v) \\x &= 1 \dots\dots (iv)\end{aligned}$$

From equation (iv), we see that $x = 1$ and back substitution gives $y = \frac{2}{3}$ and $z = \frac{1}{3}$.

Thus, $x = 1$, $y = \frac{2}{3}$ and $z = \frac{1}{3}$.

Eigenvalues and eigenvectors

If A is an $n \times n$ matrix, then a nonzero vector $\vec{x}$ in $\mathbb{R}^n$ is called an eigenvector of A if $A\vec{x}$ is a scalar multiple of $\vec{x}$, that is, $A\vec{x} = \lambda\vec{x}$ for some scalar λ . The scalar λ is called an eigenvalue of A , and $\vec{x}$ is said to be an eigenvector corresponding to λ .

Note

In general, the image of a vector $\vec{x}$ under multiplication by a square matrix A differs from $\vec{x}$ in both magnitude and direction. However, in the special case where $\vec{x}$ is an eigenvector of A ,

depending on the sign and magnitude of the eigenvalue λ corresponding to $\vec{x}$, the operation $A\vec{x} = \lambda\vec{x}$ compresses or stretches $\vec{x}$ by a factor of λ , with a reversal of direction in the case where λ is negative.

Example

The vector $\vec{x} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is an eigenvector of $A = \begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix}$ corresponding to the eigenvalue $\lambda = 3$,

$$\text{since } A\vec{x} = \begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} = 3\vec{x}$$

Geometrically, multiplication by A has stretched the vector $\vec{x}$ by a factor of 3.

Computing eigenvalues and eigenvectors

A general procedure for finding eigenvalues and eigenvectors of an $n \times n$ matrix A is by first rewriting the equation $A\vec{x} = \lambda\vec{x}$ as $A\vec{x} = \lambda I\vec{x}$ or equivalently, as $(A - \lambda I)\vec{x} = \vec{0}$.

Theorem

If A is an $n \times n$ matrix, then λ is an eigenvalue of A if and only if it satisfies the equation $\det(A - \lambda I) = 0$. This is called the characteristic equation of A.

Example

Determine the eigenvalues and associated eigenvectors of the matrix

$$A = \begin{pmatrix} -8 & -18 \\ 6 & 13 \end{pmatrix}$$

Solution

For the values of $\vec{x}$ in $(A - \lambda I)\vec{x} = \vec{0}$ to be non-zeros then $|A - \lambda I| = 0$

$$\Rightarrow A - \lambda I = \begin{pmatrix} -8 - \lambda & -18 \\ 6 & 13 - \lambda \end{pmatrix}$$

Therefore,

$$\begin{aligned}
|A - \lambda I| &= \begin{vmatrix} -8 - \lambda & -18 \\ 6 & 13 - \lambda \end{vmatrix} = 0 \\
\Rightarrow (-8 - \lambda)(13 - \lambda) + 108 &= 0 \\
\Rightarrow -104 + 8\lambda - 13\lambda + \lambda^2 + 108 &= 0 \\
\Rightarrow \lambda^2 - 5\lambda + 4 &= 0 \\
\Rightarrow (\lambda - 1)(\lambda - 4) &= 0 \\
\Rightarrow \lambda = 1 \text{ or } \lambda = 4
\end{aligned}$$

Hence, the eigenvalues are $\lambda = 1, \lambda = 4$

Eigenvector(s) associated with $\lambda = 1$:

Let $\vec{x} = \begin{pmatrix} a \\ b \end{pmatrix}$. Then

$$\begin{aligned}
(A - \lambda I)\vec{x} &= \begin{pmatrix} -9 & -18 \\ 6 & 12 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \vec{0} \\
\Rightarrow \begin{cases} -9a - 18b = 0 \\ 6a + 12b = 0 \end{cases} &\Rightarrow \begin{cases} a + 2b = 0 \\ a + 2b = 0 \end{cases} \\
\Rightarrow a &= -2b
\end{aligned}$$

Taking $a = 2 \Rightarrow b = -1$

Thus, $\vec{x} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ is the eigenvector corresponding to $\lambda = 1$

For $\lambda = 4$, and letting $\vec{x} = \begin{pmatrix} x \\ y \end{pmatrix}$, we have

$$\begin{aligned}
(A - \lambda I)\vec{x} &= \begin{pmatrix} -12 & -18 \\ 6 & 9 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \vec{0} \\
\Rightarrow \begin{cases} -12x - 18y = 0 \\ 6x + 9y = 0 \end{cases} &\Rightarrow \begin{cases} 6x + 9y = 0 \\ 6x + 9y = 0 \end{cases} \\
\Rightarrow 2x &= -3y
\end{aligned}$$

Taking $x = 3 \Rightarrow y = -2$

Hence, $\vec{x} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ is the eigenvector corresponding to $\lambda = 4$

Example

Find bases (eigenvectors) for the eigen spaces (eigenvalues) of $A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$

Solution

The characteristic equation of A is

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0, \text{ or in factored form } (\lambda - 1)(\lambda - 2)^2 = 0.$$

Hence, the distinct eigenvalues of A are $\lambda = 1$ and $\lambda = 2$

So there are two eigen spaces of A.

Let

$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

Then

$$\begin{pmatrix} -\lambda & 0 & -2 \\ 1 & 2-\lambda & 1 \\ 1 & 0 & 3-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

In the case where $\lambda = 1$ we have

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Solving this system yields $x_1 = -2s$, $x_2 = s$, $x_3 = s$

Thus the eigenvector(s) corresponding to $\lambda = 1$ are the nonzero vectors of the form

$$\vec{x} = \begin{pmatrix} -2s \\ s \\ s \end{pmatrix} = s \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$$

So that $\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$ is a basis vector for eigen space corresponding to $\lambda = 1$

Similarly, for $\lambda = 2$, we have

$$\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Solving this system using Gaussian elimination method yields $x_1 = -s$, $x_2 = t$, $x_3 = s$

Thus, the eigenvector(s) corresponding to $\lambda = 2$ are the nonzero vectors of the form

$$\vec{x} = \begin{pmatrix} -s \\ t \\ s \end{pmatrix} = \begin{pmatrix} -s \\ 0 \\ s \end{pmatrix} + \begin{pmatrix} 0 \\ t \\ 0 \end{pmatrix} = s \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Since

$$\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

are linearly independent, then they form a basis for the eigenspace corresponding to $\lambda = 2$

Theorem

If k is a positive integer, λ an eigenvalue of a matrix A and $\vec{x}$ a corresponding eigenvector, then λ^k is an eigenvalue of A^k and $\vec{x}$ is a corresponding eigenvector.

Definition

If A and B are n-square matrices, then we say that B is similar to A if there is an invertible matrix P such that $B = P^{-1}AP$.

Note

1. If matrix B is similar to matrix A then A is also similar to B.
2. Similar matrices have the same
 - (i) Determinant
 - (ii) Trace
 - (iii) Characteristic polynomial
 - (iv) Eigenvalues

Diagonalization of a square matrix

Definition

A square matrix A is said to be diagonalizable if it is similar to some diagonal matrix, that is, if there exists an invertible matrix P such that $P^{-1}AP = D$, where D is a diagonal matrix. In this case matrix P is said to diagonalize A.

Procedure for diagonalizing a matrix

Let A be an n-square matrix. Then to diagonalize A

1. Find n linearly independent eigenvectors of A. If the number of eigenvectors of A is less than n, then A is not diagonalizable
2. Form matrix P that has the eigenvectors as its column vectors
3. $P^{-1}AP$ will be diagonal and have the corresponding eigenvalues as its successive diagonal entries.

Example

Find a matrix P that diagonalizes

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$

Solution

In the example above, we found the characteristic equation of A to be $(\lambda - 1)(\lambda - 2)^2 = 0$ and eigenvectors corresponding to respective eigenvalues as

$$\lambda = 1: \vec{x}_1 = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}; \lambda = 2: \vec{x}_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \vec{x}_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

There are three basis vectors in total, so matrix

$$P = \begin{pmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Diagonalizes A. That is,

$$P^{-1}AP = \begin{pmatrix} -1 & 0 & -1 \\ 1 & 0 & 2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Example

Find a matrix P that diagonalizes

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & 5 & 2 \end{pmatrix}$$

Solution

The characteristic polynomial of A is

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 1 & 2-\lambda & 0 \\ -3 & 5 & 2-\lambda \end{vmatrix} = (1-\lambda)(2-\lambda)^2$$

Thus, the characteristic equation is

$$(1-\lambda)(2-\lambda)^2 = 0$$

The distinct eigenvalues of A are $\lambda = 1$ and $\lambda = 2$.

The eigenvectors corresponding to the respective eigenvalues are

$$\lambda = 1: \vec{x}_1 = \begin{pmatrix} \frac{1}{8} \\ \frac{-1}{8} \\ 1 \end{pmatrix}; \quad \lambda = 2: \vec{x}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Since A is a 3×3 matrix and there are only two basis vectors in total, then A is not diagonalizable.

Definition

A set of two or more vectors is said to be orthogonal if all pairs of distinct vectors in the set are orthogonal, that is, if their dot product is equal to zero. An orthogonal set in which each vector is of unit length is said to be orthonormal.

The Gram-Schmidt process

The Gram-Schmidt process is a simple algorithm for producing an orthogonal or orthonormal basis for any non-zero subspace of $\mathbb{R}^n$. Given a basis $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_p\}$ for a subspace W of $\mathbb{R}^n$, define

$$\vec{v}_1 = \vec{x}_1$$

$$\vec{v}_2 = \vec{x}_2 - \frac{\vec{x}_2 \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1$$

$$\vec{v}_3 = \vec{x}_3 - \frac{\vec{x}_3 \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1 - \frac{\vec{x}_3 \cdot \vec{v}_2}{\vec{v}_2 \cdot \vec{v}_2} \vec{v}_2$$

•

•

•

$$\vec{v}_p = \vec{x}_p - \frac{\vec{x}_p \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1 - \frac{\vec{x}_p \cdot \vec{v}_2}{\vec{v}_2 \cdot \vec{v}_2} \vec{v}_2 - \dots - \frac{\vec{x}_p \cdot \vec{v}_{p-1}}{\vec{v}_{p-1} \cdot \vec{v}_{p-1}} \vec{v}_{p-1}$$

Then, $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_p\}$ is an orthogonal basis for W.

Orthogonal matrices

Definition

A square matrix A is said to be orthogonal if its transpose is the same as its inverse, that is, if

$$A^{-1} = A^T$$

Or equivalently, if $AA^T = A^T A = I$

Example

The matrix

$$A = \begin{pmatrix} \frac{3}{7} & \frac{2}{7} & \frac{6}{7} \\ \frac{-6}{7} & \frac{3}{7} & \frac{2}{7} \\ \frac{2}{7} & \frac{6}{7} & \frac{-3}{7} \end{pmatrix}$$

Is orthogonal since

$$A^T A = \begin{pmatrix} \frac{3}{7} & \frac{-6}{7} & \frac{2}{7} \\ \frac{2}{7} & \frac{3}{7} & \frac{6}{7} \\ \frac{6}{7} & \frac{2}{7} & \frac{-3}{7} \end{pmatrix} \begin{pmatrix} \frac{3}{7} & \frac{2}{7} & \frac{6}{7} \\ \frac{-6}{7} & \frac{3}{7} & \frac{2}{7} \\ \frac{2}{7} & \frac{6}{7} & \frac{-3}{7} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Note

If A is an orthogonal matrix then

- (i) the row vectors and column vectors of A are orthonormal
- (ii) the inverse of A is orthogonal
- (iii) $\det(A) = 1$ or -1

Orthogonal diagonalization

Definition

If A and B are square matrices, then we say that A and B are orthogonally similar if there is an orthogonal matrix P such that $P^T A P = B$

If A is similar to some diagonal matrix, say $P^T A P = D$ then we say that A is orthogonally diagonalizable and that D orthogonally diagonalizes A.

Note

- (1) From $P^T A P = D$, where P is an orthogonal matrix and D is a diagonal matrix, we see that

$A = P D P^T$. Now, transposing both sides of $A = P D P^T$ and using the fact that a diagonal matrix is the same as its transpose we obtain $A^T = (P D P^T)^T = (P^T)^T D^T P^T = P D P^T = A$.

- (2) If an n-square matrix A is orthogonally diagonalizable then

- (i) A has an orthonormal set of n eigenvectors
- (ii) A is symmetric

Symmetric matrices

An $n \times n$ matrix A is said to be a symmetric matrix if $A^T = A$. Such a matrix has arbitrary main diagonal entries, but its other entries occur in pairs on opposite sides of the main diagonal.

Example

Matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & -1 & 0 \\ -1 & 5 & 8 \\ 0 & 8 & -7 \end{pmatrix}, \begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix}$$

Are symmetric, while

$$\begin{pmatrix} 1 & -3 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} 1 & -4 & 0 \\ -6 & 1 & -4 \\ 0 & 6 & 1 \end{pmatrix}, \begin{pmatrix} 5 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix}$$

Are nonsymmetric.

Diagonalization of symmetric matrices

Diagonalization of a symmetric matrix is the foundation for discussion concerning quadratic forms.

If A is a symmetric matrix, then to diagonalize it

- (a) Find a basis for each eigenspace of A
- (b) Apply the Gram-Schmidt process to each of these bases to obtain an orthonormal basis for each eigenspace
- (c) Form the matrix P whose columns are the vectors constructed in (b). This matrix will orthogonally diagonalize A, and the eigenvalues on the diagonal of $D = P^T A P$ will be in the same order as their corresponding eigenvectors in P.

Example

Find an orthogonal matrix P that diagonalizes

$$A = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{pmatrix}$$

Solution

The characteristic equation of A is

$$\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 2 & 2 \\ 2 & 4-\lambda & 2 \\ 2 & 2 & 4-\lambda \end{vmatrix} = (2-\lambda)^2(8-\lambda) = 0$$

Thus, the distinct eigenvalues of A are $\lambda = 2$ and $\lambda = 8$.

The eigenvectors for the eigenspace corresponding to $\lambda = 2$ is

$$\vec{u}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \text{ and } \vec{u}_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Applying the Gram-Schmidt process to $\{\vec{u}_1, \vec{u}_2\}$ yields the following orthonormal vectors

$$\vec{v}_1 = \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} \text{ and } \vec{v}_2 = \begin{pmatrix} \frac{-1}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{pmatrix}$$

The eigenspace corresponding to $\lambda = 8$ has

$$\vec{u}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

As a basis vector.

Normalizing $\vec{u}_3$ gives

$$\vec{v}_3 = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}$$

Finally, using $\vec{v}_1$, $\vec{v}_2$ and $\vec{v}_3$ as column vectors, we obtain

$$P = \begin{pmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

Which orthogonally diagonalizes A, that is,

$$P^T A P = \begin{pmatrix} \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{-1}{\sqrt{6}} & \frac{-1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{pmatrix} \begin{pmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix}$$

Quadratic forms

Quadratic forms occur frequently in applications of linear algebra to engineering (in design criteria and optimization) and signal processing (as output noise power). They also arise, for instance, in physics (as potential and kinetic energy), differential geometry (as normal curvature of surfaces), economics (as utility functions) and statistics (in confidence ellipsoids).

A quadratic form in $\mathbb{R}^n$ is a function Q defined on $\mathbb{R}^n$ whose value at a vector $\vec{x}$ in $\mathbb{R}^n$ can be computed by an expression of the form $Q(\vec{x}) = \vec{x}^T A \vec{x}$ where A is an $n \times n$ symmetric matrix.

Matrix A is called the matrix of the quadratic form.

The simplest example of a nonzero quadratic form is $Q(\vec{x}) = \vec{x}^T I \vec{x} = |\vec{x}|^2$

Examples

1. Let $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$. Compute $\vec{x}^T A \vec{x}$ for the following matrices

(a) $A = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}$

(b) $A = \begin{pmatrix} 3 & -2 \\ -2 & 7 \end{pmatrix}$

Solution

$$\begin{aligned} \vec{x}^T A \vec{x} &= (x_1 \quad x_2) \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ \text{(a)} \quad &= (x_1 \quad x_2) \begin{pmatrix} 4x_1 \\ 3x_2 \end{pmatrix} \\ &= 4x_1^2 + 3x_2^2 \end{aligned}$$

$$\begin{aligned} \vec{x}^T A \vec{x} &= (x_1 \quad x_2) \begin{pmatrix} 3 & -2 \\ -2 & 7 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ &= (x_1 \quad x_2) \begin{pmatrix} 3x_1 - 2x_2 \\ -2x_1 + 7x_2 \end{pmatrix} \\ \text{(b)} \quad &= x_1(3x_1 - 2x_2) + x_2(-2x_1 + 7x_2) \\ &= 3x_1^2 - 2x_1x_2 - 2x_1x_2 + 7x_2^2 \\ &= 3x_1^2 - 4x_1x_2 + 7x_2^2 \end{aligned}$$

2. For $\vec{x}$ in $\mathbb{R}^3$, let $Q(\vec{x}) = 5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3$. Write this quadratic form as

$$\vec{x}^T A \vec{x}$$

Solution

The coefficients of x_1^2 , x_2^2 , x_3^2 go on to the diagonal of A. To make A symmetric, the coefficient of $x_i x_j$ for $i \neq j$ must be split evenly between the (i, j)- and (j, i)- entries in A. The coefficient of $x_1 x_3$ is 0. Thus,

$$Q(\vec{x}) = \vec{x}^T A \vec{x} = \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{pmatrix} 5 & \frac{-1}{2} & 0 \\ \frac{-1}{2} & 3 & 4 \\ 0 & 4 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

3. Let $Q(\vec{x}) = x_1^2 - 8x_1x_2 - 5x_2^2$. Compute the value of $Q(\vec{x})$ for

$$\vec{x} = \begin{pmatrix} -3 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \end{pmatrix} \text{ and } \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

Solution

$$Q(-3, 1) = (-3)^2 - 8(-3)(1) - 5(1)^2 = 28$$

$$Q(2, -2) = (2)^2 - 8(2)(-2) - 5(-2)^2 = 16$$

$$Q(1, -3) = (1)^2 - 8(1)(-3) - 5(-3)^2 = -20$$

Note

1. There are three important kinds of problems that occur in applications of quadratic forms:

- If $\vec{x}^T A \vec{x}$ is a quadratic form on $\mathbb{R}^2$ or $\mathbb{R}^3$, what kind of curve or surface is represented by the equation $\vec{x}^T A \vec{x} = k$?
- If $\vec{x}^T A \vec{x}$ is a quadratic form on $\mathbb{R}^n$, what conditions must A satisfy for $\vec{x}^T A \vec{x}$ to have positive values for $\vec{x} \neq \vec{0}$?
- If $\vec{x}^T A \vec{x}$ is a quadratic form on $\mathbb{R}^n$, what are its maximum and minimum values if $\vec{x}$ is constrained to satisfy $|\vec{x}| = 1$?

2. In some cases, quadratic forms are easier to use when they have no cross-product terms, that is, when the matrix of the quadratic form is a diagonal matrix. Fortunately, the cross-product term can be eliminated by making a suitable change of variable.

Change of variable in a quadratic form

If $\vec{x}$ represents a variable vector in $\mathbb{R}^n$, then a change of variable is an equation of the form $\vec{x} = P\vec{y}$, or equivalently, $\vec{y} = P^{-1}\vec{x}$, where P is an invertible matrix and $\vec{y}$ is a new variable

vector in $\mathbb{R}^n$. Here $\vec{y}$ is the coordinate vector of $\vec{x}$ relative to the basis of $\mathbb{R}^n$ determined by the columns of P.

If the change of variable is made in a quadratic form $\vec{x}^T A \vec{x}$, then

$$\begin{aligned}\vec{x}^T A \vec{x} &= (P\vec{y})^T A (P\vec{y}) \\ &= \vec{y}^T P^T A P \vec{y} \\ &= \vec{y}^T (P^T A P) \vec{y}\end{aligned}$$

And the new matrix of the quadratic form is $P^T A P$. If P orthogonally diagonalizes A, then $P^T = P^{-1}$ and $P^T A P = P^{-1} A P = D$. The matrix of the new quadratic form is then a diagonal matrix.

Example

Make a change of variable that transforms the quadratic form $Q(\vec{x}) = x_1^2 - 8x_1x_2 - 5x_2^2$ into a quadratic form with no cross-product term.

Solution

The matrix of the quadratic form is

$$A = \begin{pmatrix} 1 & -4 \\ -4 & -5 \end{pmatrix}$$

The first step is to orthogonally diagonalize A. Its eigenvalues turn out to be $\lambda = 3$ and $\lambda = -7$. Associated eigenvectors are

$$\lambda = 3: \begin{pmatrix} \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{pmatrix}, \quad \lambda = -7: \begin{pmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix}$$

These vectors are automatically orthogonal (because they correspond to distinct eigenvalues) and so provide an orthonormal basis for $\mathbb{R}^2$. Let

$$P = \begin{pmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{-1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & -7 \end{pmatrix}.$$

Then, $A = PDP^{-1}$ and $D = P^{-1}AP = P^T AP$.

A suitable change of variable is $\vec{x} = P\vec{y}$, where $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $\vec{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$

Then,

$$\begin{aligned} x_1^2 - 8x_1x_2 - 5x_2^2 &= \vec{x}^T A \vec{x} = (P\vec{y})^T A (P\vec{y}) \\ &= \vec{y}^T P^T A P \vec{y} = \vec{y}^T D \vec{y} \\ &= 3y_1^2 - 7y_2^2 \end{aligned}$$

Note

To illustrate the meaning of the equality of quadratic forms, we can compute $Q(\vec{x})$ for

$\vec{x} = (2, -2)$ using the new quadratic form. First, since $\vec{x} = P\vec{y}$, we have $\vec{y} = P^{-1}\vec{x} = P^T \vec{x}$

So

$$\vec{y} = P^T \vec{x} = \begin{pmatrix} \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix} \begin{pmatrix} 2 \\ -2 \end{pmatrix} = \begin{pmatrix} \frac{6}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{pmatrix}$$

Hence ,

$$\begin{aligned} 3y_1^2 - 7y_2^2 &= 3\left(\frac{6}{\sqrt{5}}\right)^2 - 7\left(\frac{-2}{\sqrt{5}}\right)^2 \\ &= 3\left(\frac{36}{5}\right) - 7\left(\frac{4}{5}\right) \\ &= \frac{80}{5} \\ &= 16 \end{aligned}$$

This is the same as the value of $Q(\vec{x})$ in the previous example when $\vec{x} = (2, -2)$.

Classifying quadratic forms

1. A quadratic form is
 - (i) Positive definite if $Q(\vec{x}) > 0$ for all $\vec{x} \neq \vec{0}$
 - (ii) Negative definite if $Q(\vec{x}) < 0$ for all $\vec{x} \neq \vec{0}$
 - (iii) Indefinite if $Q(\vec{x})$ assumes both positive and negative values
2. Let A be an $n \times n$ symmetric matrix. Then a quadratic form $\vec{x}^T A \vec{x}$ is
 - (i) Positive definite iff the eigenvalues of A are all positive
 - (ii) Negative definite iff the eigenvalues of A are all negative or
 - (iii) Indefinite iff A has both positive and negative eigenvalues

Example

Is $Q(\vec{x}) = 3x_1^2 + 2x_2^2 + x_3^2 + 4x_1x_2 + 4x_2x_3$ positive definite?

Solution

The quadratic form looks positive definite because of all the plus signs. But, the matrix of the form is

$$A = \begin{pmatrix} 3 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 1 \end{pmatrix}$$

And the eigenvalues of A turn out to be 5, 2, and -1. So Q is an indefinite quadratic form, not positive definite.

Optimization using quadratic forms

Quadratic forms arise in various problems in which the maximum or minimum value of some quantity is required. We consider the problem of finding the maximum and minimum values of a quadratic form $\vec{x}^T A \vec{x}$ subject to the constraint $|\vec{x}| = 1$.

Theorem (constrained extremum theorem)

Let A be an $n \times n$ symmetric matrix whose eigenvalues in order of decreasing size are

$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. Then the

- (a) quadratic form $\vec{x}^T A \vec{x}$ attains a maximum value and a minimum value on the set of vectors for which $|\vec{x}| = 1$
- (b) maximum value attained in part (a) occurs at a unit vector corresponding to the eigenvalue λ_1
- (c) minimum value attained in part (a) occurs at a unit vector corresponding to the eigenvalue λ_n

Example Finding constrained extrema

Compute the maximum and minimum values of the quadratic form

$$z = 5x^2 + 5y^2 + 4xy$$

Subject to the constraint $x^2 + y^2 = 1$

Solution

The quadratic form can be expressed in matrix notation as

$$z = 5x^2 + 5y^2 + 4xy = \vec{x}^T A \vec{x} = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

The eigenvalues of A are $\lambda_1 = 7$ and $\lambda_2 = 3$ and the corresponding eigenvectors are

$$\lambda_1 = 7 : \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \lambda_2 = 3 : \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Normalizing these eigenvectors yields

$$\lambda_1 = 7 : \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \quad \lambda_2 = 3 : \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

Thus, the constrained extrema are

$$\text{Constrained maximum is } z = 7 \text{ at } (x, y) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

Constrained minimum is $z = 3$ at $(x, y) = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

Remark

Since the negatives of the eigenvectors above are also unit eigenvectors, they too produce the maximum and minimum values of z ; that is, the constrained maximum $z = 7$ also occurs at point $(x, y) = \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$ and the constrained minimum $z = 3$ at $(x, y) = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$.

Example

A rectangle is to be inscribed in an ellipse $4x^2 + 9y^2 = 36$. Use eigenvalue methods to find nonnegative values of x and y that produce the inscribed rectangle with maximum area.

Solution

The area z of the inscribed rectangle is given by $z = 4xy$, so the problem is to maximize the quadratic form $z = 4xy$ subject to the constraint $4x^2 + 9y^2 = 36$. In this problem, the constraint equation is an ellipse rather than a unit circle as required by the theorem above, but we can remedy this problem by rewriting the constraint as

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$$

And defining new variables, x_1 and y_1 , by the equations

$$x = 3x_1 \text{ and } y = 2y_1.$$

This enables us to reformulate the problem as:

Maximize $z = 4xy = 24x_1y_1$ subject to the constraint $x_1^2 + y_1^2 = 1$.

To solve this, we write the quadratic form $z = 24x_1y_1$ as

$$z = \vec{x}^T A \vec{x} = \begin{pmatrix} x_1 & y_1 \end{pmatrix} \begin{pmatrix} 0 & 12 \\ 12 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

The largest eigenvalue for A is $\lambda = 12$ and the only corresponding unit eigenvector with nonnegative entries is

$$\vec{x} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

Thus, the maximum area is $z = 12$, and this occurs when

$$x = 3x_1 = \frac{3}{\sqrt{2}} \text{ and } y = 2y_1 = \frac{2}{\sqrt{2}}.$$

Sample Examination Questions

- Suppose U, V are subspaces of $\mathbb{R}^n$. Check whether or not
 - $U \cap V$ is a vector subspace of $\mathbb{R}^n$ **(2 marks)**
 - $U \cup V$ is a vector subspace of $\mathbb{R}^n$ **(2 marks)**
- Check whether or not $(1, 1, -1, 1)$ is a linear combination of $(1, 1, 0, 0), (1, 0, 0, 1), (1, 0, 1, 0), (0, 0, 1, 1)$ **(3 marks)**
- Let $V = \{(x, y, z) : x - 2y + z = 3\}$. Check whether or not V is a subspace of $\mathbb{R}^3$
- Check for linear dependence in $(3, 4, 5), (1, 2, 0), (0, 1, 2)$ **(3 marks)**
- Determine the dimension of the vector space spanned by the vectors $\vec{u} = (2, -1, 3), \vec{v} = (4, 0, 6)$ and $\vec{w} = (8, -2, -3)$. **(5 marks)**
- A Television Room has 24 tables; x tables each seating 4, y tables each seating 6 and z tables each seating 10. The total seating capacity in the room is 148. For a special second year's student gathering, one-half of the x tables, one-fourth of the y tables and one-third of the z tables for a total of 9 tables was used. Find how many of each tables are in the room. **(10 marks)**